Physical and Chemical Change Revision Quiz

1. Identify the state(s) of matter in which the particles take the shape of the container.
   a. Liquids and gases
   b. Gases only
   c. Liquids only
   d. Liquids and solids

2. Define diffusion
   a. Diffusion is when particles in a liquid or gas mix evenly through random motion.
   b. Diffusion is when particles can be compressed.
   c. Diffusion is when particles cannot be compressed.
   d. Diffusion is when particles have a large amount of space between them.

3. Identify what happens when a substance dissolves
   a. The particles of a substance disappear in water.
   b. The particles chemically react with water.
   c. The substances changes state.
   d. The particles of the substance are distributed evenly through particles of another substance.

4. Recall the effect that heat has on the energy of particles.
   a. Heat causes the particles to change shape.
   b. Heat causes the particles to change state.
   c. Heat causes the particles to move faster.
   d. Heat causes the particles to melt.

5. Identify the name of the process that occurs when a solid carbon dioxide (dry ice) is converted to gas.
   a. Evaporation.
   b. Boiling
   c. Sublimation
   d. Melting.

6. Identify the correct statement(s) for the particle theory of matter.
    a. The particle theory of matter states that all matter is made up of particles.
    b. All the answers are correct
    c. The particle theory of matter states that particles of matter are constantly moving.
    d. The particle theory of matter explains the behaviour of solids, liquids and gases.

7. Explain why metals expand when heated.
    a. The particles gain energy and move further apart.
    b. The particles gain energy and get larger.
    c. The particles lose energy and get larger.
    d. The particles lose energy and move further apart.

8. There are ways to tell if a chemical reaction has occurred. Identify which of the following is NOT one of these
    a. The reaction vessel becomes hotter
    b. There is a change of state
    c. A gas is produced
    d. A precipitate forms

9. Identify which of these reactions is endothermic
    a. A sparkler producing coloured lights.
    b. Burning a lump of coal
    c. Photosynthesis
    d. A lit candle of a birthday cake

10. Identify which of the following is not a sign of chemical change.
    a. A solid precipitate or gas is produced.
    b. There is a permanent colour change
    c. Energy is produced or absorbed
    d. A mixture forms

11. Identify which of the following is a different substance (contains different types of particles) to the others
    a. Ice
    b. Water
    c. Air
    d. Steam

12. Identify which of the following are difficult to compress.
    a. Only solids
    b. Solids and liquids
    c. Liquids and gases
    d. Solids and gases

13. Identify the correct statement. The melting point of a substance is the:
    a. Part of an object that first turns to liquid.
    b. Temperature at which a substance begins to melt
    c. Time taken for a substance to fully melt
    d. Place where it is when it melts.

14. Identify the correct statement. Evaporation is:
    a. The same as solidification
    b. Something that happens to gases
    c. When small bubbles begin to appear in a liquid.
    d. Another term for vaporisation.

15. Identify the correct statement. Dew on the grass in the morning after a cold night is the result of:
    a. Sublimation
    b. Diffusion
    c. Evaporation
    d. Condensation

16. Determine which of the following is the least expensive form of rust protection.
    a. Alloying
    b. Galvanization
    c. Chrome plating
    d. Painting.

17. Identify which of the following are units of density.
    a. Kg/m^2
    b. M^3/kg
    c. Kg/m^3
    d. m/kg^3

18. Identify the correct statement about density.
    a. Lead has a lower density than water.
    b. A large piece of copper is denser than a small piece of copper.
    c. A solid has a higher density than any type of liquid.
    d. A copper coin has the same density as a copper cube.

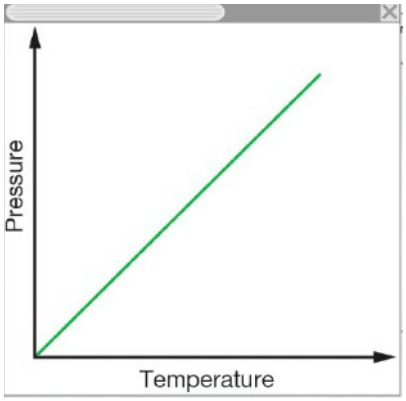
19. Identify the correct statement. An object will float if its density is:
    a. Lower than that of the liquid it is in.
    b. Less than 1 g/cm^3
    c. The same as that of the liquid it is in.
    d. Higher than that of the liquid it is in.

20. Identify where the heat given off by an exothermic reaction comes from.
    a. Atoms that are destroyed during the reaction and converted to energy.
    b. The collisions between atoms and molecules.
    c. The energy stored in the chemical bonds of the reactants
    d. The surroundings (environment)

21. Identify the products in the chemical reaction: nitric acid + lithium hydroxide → lithium nitrate + water
    a. Nitric acid and water
    b. Nitric acid and lithium hydroxide
    c. Lithium hydroxide and lithium nitrate
    d. Lithium nitrate and water.

22. Determine the formula equation for the word equation: copper sulphate + barium chloride → copper chloride + barium sulphate.
    a. $CuSO_4 + Ba(OH)_2 \rightarrow Cu(OH)_2 + BaSO_4$
    b. $CuSO_4 + BaCl_2 \rightarrow CuCl_2 + BaSO_4$
    c. $H_2SO_4 + CuCO_3 \rightarrow CuSO_4 + CO_2 + H_2O$
    d. $CuCO_3 + BaCl_2 \rightarrow CuCl_2 + BaCO_3$

23. Deduce from this graph the relationship between pressure and temperature of a gas in a solid container.
    a. As the temperature increase, the pressure decreases.
    b. As the temperature increases, the pressure stays constant.
    c. As temperature increases, the pressure increases
    d. As the temperature decreases, the pressure increases.



24. Read the data in the table below and identify the correct statement:
    a. Metals expand less than non-metals
    b. Aluminium expands more than steel
    c. Metals expand more than non-metals
    d. Aluminium expands less than brass

1 metre length expansion table

Temperature	10°C	100°C	1000°C
	Expansion amount (mm)		
Invar (nickel-steel mixture)	0.001	0.01	0.1
Wood (oak)	0.003	0.03	0.3
Pyrex	0.003	0.03	0.3
Glass	0.009	0.09	0.9
Platinum	0.009	0.09	0.9
Steel	0.011	0.11	1.1
Concrete	0.011	0.11	1.1
Iron	0.012	0.12	1.2
Brass	0.019	0.19	1.9
Aluminium	0.025	0.25	2.5

25. Explain what is happening in this diagram.
    a. The atoms in the reacts are being destroyed and the products are being created.
    b. The atoms in the reactants are rearranging to form the products
    c. This is a physical change because no new atoms have been formed.
    d. The atoms in the products are rearranging to form the reactants.

